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Discussion Agenda 05/09/2013 (Rockstar + yt in AGORA)

To investigate halo catalogues, halo merger histories, subhalo mass function and others, adopting a reliable, well-tested halo finder is a key for the success of the Project. As a first step, the Flagship paper will present the subhalo mass function of the [proof-of-concept DM-only Flagship run](#) ($M_{\text{vir}} \sim 10^{11} M_{\text{sun}}$ @ $z=0$). To achieve this goal, Working Groups I and IV are trying to employ the [Rockstar halo finder](#) (Behroozi et al. 2013) via the yt common analysis platform. The document below describes such effort (therefore naturally overlaps with [this page](#)).

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Participants of WGs I and IV discussed some of the items below on May 9. Two major issues were:

- [1] Status and effectiveness of the newest Rockstar (Rockstar-galaxies) to identify peaks in multiple-mass zoom-in simulations like AGORA
- [2] Inclusion of such new Rockstar-galaxies into yt, the common analysis platform for AGORA

**PLEASE READ THE IMPORTANT QUESTIONS AND ACTION ITEMS BELOW
IN RED BOLDFACE !**

1. Halo Finder in AGORA Project

(1) Question 1:

- Do we all agree to use the Rockstar halo finder for the Flagship paper? Should the "official" halo finder of the AGORA Project be announced in the Flagship paper?

[Proposed Policy]

- We will employ the Rockstar halo finder to analyze the [proof-of-concept DM-only run](#) for the use in the Flagship paper and the companion paper to introduce the ICs in detail (Hahn et al. 2013). However, the choice of halo finder for subsequent science-oriented papers will be left entirely open.

- Regardless of your choice of a halo finder, measuring halos or building halo merger trees may be done in completely different and independent modules. These are in principle all separable choices. (For example, yt supports these functionalities; see [halo profiling](#), [halo mass function](#), and [halo merger tree](#) in yt documentations thanks to Stephen Skory and others.)

(2) Question 2:

- For future hydrodynamics simulations in the AGORA Project, how do we compare the results of halo finders across particle-based and grid-based codes? (Halo finders are run on DM+star+gas particles in particle-based codes, but frequently only on DM particles in grid-based codes.)

[Proposed Policy]

- To compare apples to apples, we propose to consider discretizing the gas grids into particles for AMR codes. This is certainly beyond the scope of the Flagship paper, but will need to be carefully examined in subsequent projects.

2. Rockstar in AGORA Project

(1) Status of Rockstar

Rockstar v.0.99.6	linked to yt as of May 2013
Rockstar v.0.99.9	extensively used and tested with DM-only simulations, does not support multiple-mass, publicly available version via Google code as of May 2013, comes with a whole package including merger tree and halo catalogue production
Rockstar-galaxies	recently updated version for multiple-mass simulations, currently in test mode

(2) Strategy to test Rockstar-galaxies and make it ready for the Flagship paper

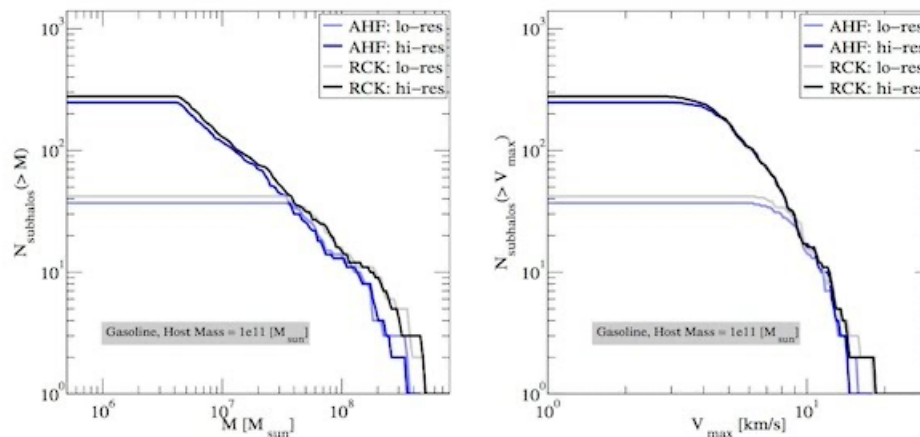
[Action Item #1]

– Two groups, UCSC (led by Pillepich and Shen) and UNLV (led by Nagamine), will first test Rockstar-galaxies with the **proof-of-concept DM-only run** performed on GASOLINE and GADGET-3, respectively. Then they will test Rockstar-galaxies with their available hydrodynamics simulations (Eris, etc) to characterize its performance between DM-only and gas simulations. In two weeks, we will discuss the test results; see the next Section.

3. Rockstar Test Results (Preliminary)

(1) Rockstar v.0.99.9 and Amiga Halo Finder (AHF, Knollmann & Knebe 2009) compared on the **proof-of-concept DM-only runs** in GASOLINE, performed by Annalisa Pillepich

- Global subhalos statistics are all right ($N_{\min} = 100$)
- Some implicit choices in Rockstar will need to be discussed (e.g. definition of total mass, truncation of range of radii for profiling, etc.)



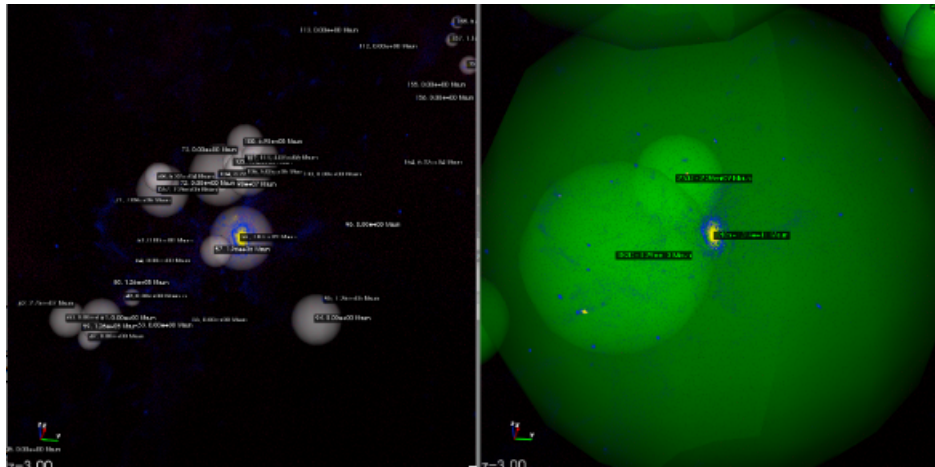
(2) Rockstar v.0.99.9 and AHF compared on a DM-only run of a MW-like galaxy in GASOLINE, performed by Annalisa Pillepich

– (plots coming soon!)

(3) Rockstar-galaxies and Spline Kernel Interpolative Denmax (SKID, Stadel 2001) compared on a multiple-mass hydrodynamics simulation in

GADGET-3, performed by Robert Thompson

- [Click](#) to see the larger image



4. Rockstar via *yt* Common Analysis Platform

(1) Rockstar in *yt*

Description of Rockstar analysis module in <i>yt</i>	See this <i>yt</i> project webpage (thanks to Matthew Turk, Christopher Moody, and Stephen Skory)
Source code of Rockstar analysis module in <i>yt</i>	See this link in Bitbucket
Recently discussed interface change	See this link

(2) Proposed Rockstar API changes

- Add `particle_type`, `particle_mass` for multiple-mass simulations
- Add `particle_softening-length` (or, `cell-size`) for GADGET-3-AFS (adaptive force softening) or AMR codes

[Action Item #2]

– As soon as Peter Behroozi finalizes the new API for Rockstar-galaxies, the Rockstar version linked in *yt* and its relevant callback functions will be updated accordingly. Then participating code groups should be able to test the newest Rockstar on the [proof-of-concept DM-only run](#) within the *yt* platform.

5. Miscellaneous: Halo Finder Resolution Test

(1) Subhalo mass function of the [proof-of-concept DM-only runs](#) in GASOLINE (RUN-1 and RUN-2; i.e. low- and high-resolution) analyzed with the Amiga Halo Finder (AHF)

- Convergence is decent ($N_{\min} = 20$)
- See also the test above in Section 3.1

